

Research Article

AN ENERGY AND ECOLOGICAL ASSESSMENT OF THE PRIOR AND POST-HYBRID RENOVATION STATE OF MULTI-DWELLING BUILDING IN THE ALGERIAN CITY OF SKIKDA

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ABSTRACT

In the context of Algerian energy and environmental concerns and its aspirations for energy transition, this article examines the pre- and post-state of a multi-family residential building in the city of Skikda, in northern Algeria, utilizing the UBEK reduced-order approach, alongside QGIS, CEA and METEONORM tools. Primarily focusing on the building's solar exposure, energy aspects, and environmental considerations. Six hybrid renovation scenarios were considered then, and assessed for their contribution for energy savings and ecological improvement. The main results revealed that 87% of annual energy requirements were related to heating and hot water, which are supplied by a natural gas system, a source of various types of greenhouse gases, which suggested the use of triple glazing, with or without two selective low-emissivity coatings only, or be combined by changing hot-water supply system to an electric one, alongside the installation of the hybrid PVT panels on the south façade. This triple glazing was joint in another scenario by integrating the same type of panels on the roof alongside a seasonal thermal energy storage system. This latter scenario being the best in terms of energy and ecology, and making the building highly performant in both aspects.

Keywords: *Electric boilers, Energy transition, High-performance glazing, Hybrid PVT, Renovation scenarios, UBEK approach*

Introduction

Several global ecological and energy issues are currently in a state of emergency, including global warming, a significant increase in the earth's average temperature, and the depletion

of certain natural resources, particularly in the energy industry in terms of the use and manufacture of fossil fuels (Larrère, 2009). Also, a widespread recognition of the end of the fossil fuel era, the global oil market has experienced

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massive disruption in recent years, placing oil-producing countries such as Algeria in financial jeopardy, with some fossil fuels predicted to run out by 2030 (Bekhtache, 2012), in addition to the rapid demographic growth that this country has seen over the last six decades, and a remarkable development in the construction industry, particularly in the residential sector, which accounts for 43% of overall energy consumption at national level, according to 2015 data (Ahmari & Korichi, 2019), and is, in fact, the most energy-intensive sector in Algeria. Skikda, a Mediterranean northeastern city of Algeria, and as a part of this country, mirrors its reality and faces exactly the same ecological as well as energy issues (Hadeif, 2020). The revolution in the building sector in Skikda, as in the whole of the Algerian country has pushed aside the nuanced specificities of the diversified climatic context, ignoring and seldom implementing traditional remedies developed through generations to improve their environments. This schism between the built environment and its framework quickly demonstrated itself in the creation of a place to call home devoid of even the most basic comforts. This problem has been exacerbated by the use of cooling, heating, along with artificial lighting, but this has resulted in an excessive and unreasonable utilization of energy resources (Latrecche & Sriti, 2018).

All of the above illustrates the state of ecological and energy vulnerability on an international scale, and even in Algeria as in Skikda; and also shows the large contribution of the residential sector to the enormous consumption of energy based on sources of fossil origin, which means that this sector participates significantly in the disruption of the environmental and climatic state as well, guiding research into the influence of hybrid physical interventions on the urban stock through renovation, as part of the energy transition. Energy renovation is a physical intervention that involves doing work on an existing building with the goal of improving its energy performance for environmentally friendly construction. This operation considers both the reduction of energy use, such as gas or electricity, and the rising of the degree of comfort of the inhabitants. It is considered to be one of the most important

approaches for ensuring the long-term sustainability of built environments at low cost, while improving quality of life (Vie-publique website, 2021). Energy renovation of buildings could be achieved through passive interventions on the building envelope, contributing to the reduction of energy needs, or through active measures by the integration of renewable energy installations, or even through hybrid interventions by coupling the mentioned two types. Many earlier researches where about the impact of this different measures of ameliorations on the energy efficiency and ecology footprint. As examples of the first type, (Stovall et al., 2007.) in its experimental study on typical houses in many places in the United States, entered wall renovations, involving the replacement of cladding, the addition of insulation under cladding as well as airtightness methods for replacement windows. In addition to the research of (Santamouris, 2014), at the Greek city of Athens, in which the roof was supplied by vegetation layers. Many other researches where about the second type of interventions, for instance, and corresponding to (Mauri, 2016) study, in the headquarters of the Province of Agrigento in Italy, the renovation procedure was through the installation of photovoltaic panels and a geothermal heat pump. The third type has also been the subject of numerous studies, such as that by (Corrado et al., 2014) on an apartment building in Italy, through the insulation of walls, as well as upper and lower floors; in addition to the integration of low-emissivity windows and solar shading devices, alongside the installation of solar thermal systems and photovoltaic panels. Furthermore, (Akande et al., 2014), who also investigated the application of passive retrofit through wall insulation and the installation of double-glazed windows, as well as the integration of photovoltaic panels, biomass with geothermal heat pumps, and ventilation with heat recovery, as part of their research on a public heritage building in the Belgian city of Brussels. These studies, in conjunction with others on the different technological methods of passive and active improvement on an architectural scale, investigated their effects in terms of reduced energy usage and greenhouse gas emissions, like (Ferrara et al., 2014; Nabinger &

Persily, 2011), together with others on an urban scale, as example (Civiero et al., 2021; Jepsen et al., 2022), showed the efficiency of these different measures of improvement in terms of energy and ecology. Certain other researches were on the economic and social aspect of renovation (Stenberg, 2018; Mikulić et al., 2020). The aim of this study is to evaluate both of the energy consumption and carbon emissions of a multi-family building built in 1978 in the Algerian Mediterranean town of Skikda, as a new context and sample, and then to carry out a similar evaluation of post-renovation scenarios following the implementation of new types of improvement measures and combinations, mainly the use of triple glazing, which was coupled with photovoltaic-thermal system (or hybrid PVT, which is a system that comprises a photovoltaic module for electrical energy conversion along with a solar system designed for highly efficient thermal energy conversion, all housed within one frame) in some cases, where the quantity and position of this latter was modified according to different scenarios, where there was also an integration of the storage system and a change in the supply system. The results help at the renovation planning stage to make the best choice and decisions, based on the energy and ecological state of the building before and after renovation, by understanding the effect of the hybrid mentioned measures, as well as selecting the most appropriate position of hybrid PVT panels, and the most suitable combination of several interventions, thanks to the combination of different simulation tools, using a new kind of approach.

Methods and materials

Methodology

Research workflow

The methodology of this present study is based on the numerical modeling and simulation method, by calibrating the simulated results on the basis of real energy consumption data, which was gathered from the local commercial agency of SONEGAS (National Company for Electricity and Gas) in Skikda.

Geometric modeling was performed using the QGIS tool, which generated a "Shapefile" containing geometrical as well as geographic

information about the case study and its neighborhood, which was subsequently imported within the CEA simulation tool, before adding climatic data for the context of the city of Skikda based on the EPW file generated by the METEONORM 8.0 meteorological program. CEA's simulation findings were calibrated and validated through the real energy consumption data, which was analyzed and assessed energetically and ecologically, before any measure of amelioration.

UBEM (Urban Building Energy Modeling) bottom-up reduced-order approach, which is recent and its first development appearance not exceeding 8 years- is the main applied approach in this study, it takes into account the surrounding urban context, as well as the geographical and climatic characteristics of the site, unlike approaches that consider only the building envelope and its interior properties. This approach is frequently used for rapid assessment of building energy and ecology performance.

Several scenarios were launched afterwards, where:

Post-renovation scenario 1 (P.R.1): Replacement of simple glazing by triple glazing.

Post-renovation scenario 2 (P.R.2): Replacement of simple glazing by triple glazing with two selective low-emissivity coatings.

Post-renovation scenario 3 (P.R.3): Replacement of simple glazing by triple glazing with two selective low-emissivity coatings + Integration of hybrid PVT system where the panels are on roof, with surface of 192 m² (the surfaces which receive annual accumulated solar radiation of above 1620 [kWh/m²/yr.]).

Post-renovation scenario 4 (P.R.4): Replacement of simple glazing by triple glazing with two selective low-emissivity coatings + Integration of hybrid PVT system where the panels are on southern vertical facade, with surface of 192 m² (the surfaces which receive annual accumulated solar radiation of above 1190 [kWh/m²/yr.]).

Post-renovation scenario 5 (P.R.5): Replacement of simple glazing by triple glazing with two selective low-emissivity coatings + Integration of hybrid PVT system where the panels are on southern vertical facade, with surface of 344 m² (the surfaces which receive

annual accumulated solar radiation of above 1150 [kWh/m²/yr.]).

Post-renovation scenario 6 (P.R.6): Replacement of simple glazing by triple glazing with two selective low-emissivity coatings + Replacement of hot water supply system (from naturel gas fired boilers to electrical boilers) + Integration of hybrid PVT system where the panels are on southern vertical facade, with surface of 344 m² (the surfaces which receive annual accumulated solar radiation of above 1150 [kWh/m²/yr.]).

Because of the technical ease of windows types changing compared to other types of physical intervention, and to compare the energy savings of triple glazing windows, with then triple glazing windows associated by two selective low-emissivity coatings, the improvement measures in the various scenarios were by changing simple glazing by these two mentioned types in two different scenarios.

The integration of solar photovoltaic-thermal panels on the roof in one scenario, and the same surface of the panels on the southern facade in another, and then an integration of a larger area of photovoltaic-thermal panels on the southern facade was to get an idea on the impact of the changing of panels placement (orientation) and quantity; and finally, the changing of hot water supply system from one based on natural gas to an electrical one, was in the final scenario to confirm the importance of the good combination between measures. The panels on the roof are oriented horizontally and attached directly to the roof surface, while the panels on the south facade are oriented vertically and attached directly to the surface of the south facade. The simulation results of these scenarios were also analyzed and evaluated from an energy and environmental point of view.

Presentation of the city context and the case study

Skikda, a city which is located in the north-east of Algeria, has an altitude ranging from 0 m to nearly 150 m above sea level with a population density of 3896 people per square kilometer (ANIREF, 2020).

In accordance with measurements of daily temperatures as well as weather conditions during the period within 1991 to 2020 (Site Climats et voyages, 2020), the geographic area is affiliated to the temperate Mediterranean climate area, which is usually humid, alongside summers that may be hot but wintertime that is generally not severe, with a typical lowest temperature in the coldest period of the year (February) of 9°C and a mean highest temperature in the summer's hottest month (August) of 30°C. The significant annual temperature variance in Skikda requires specific strategies to ensure adequate thermal comfort inside buildings. Prior investigations indicate that Skikda's multi-family dwellings are in a degraded state because of historical and meteorological factors, and that the lack of housing resulted in rapid urbanization without the application of bioclimatic principles (Brighet, 2018). All of these purposes led to the creation of residents that are not provided with sufficient comfort and do not meet current energy conservation standards (Kassis, 2012). Consequently, these buildings have to combat both daily and annual weather fluctuations, necessitating the significant use of artificial methods which are polluting and energy-intensive (Boulkenafet, 2014).

The study case is a multi-dwelling housing comprising 30 apartments distributed across 10 floors. It is situated in a 700-unit housing estate at the heart of the city. (Figure 1) precisely depicts its location. Its selection was based on its age and size, making it an effective sample to reflect energy consumption patterns in Skikda, with the presence of 30 families.

The different technical and administrative information pertaining to the case studies is assembled from the numerous agencies affiliated with the Wilaya of Skikda, including the city municipality hall, the OPGI (which refers to Real Estate Promotion & Management Office), and particular design offices of the key contractors. The information in Table (1) relates to the construction year alongside architectural system, geometry, height, and thermal transmittance (U-value). This information was used to model the building envelope and insert its various characteristics using the simulation tools provided.

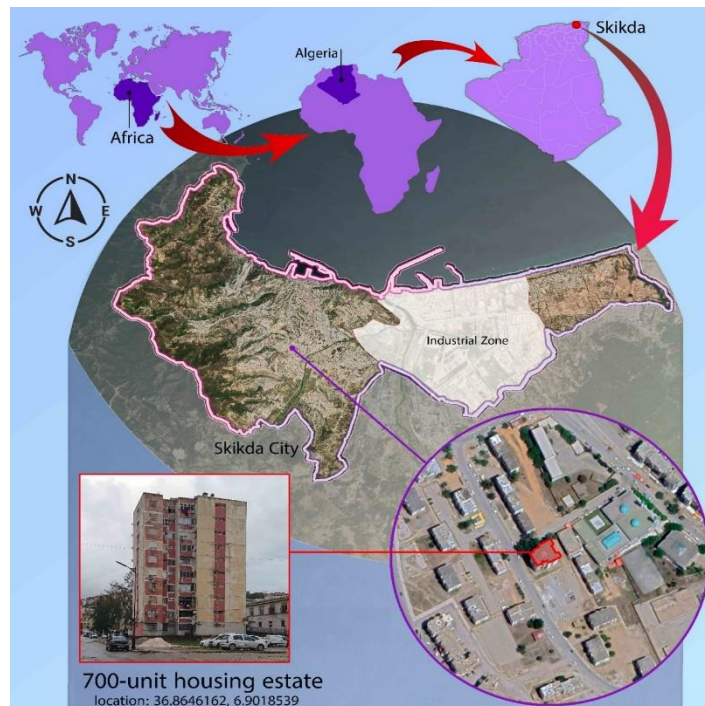


Fig. 1. Designation of study case position in relation to the world (By the authors).

Table 1. Main properties and data related to the archetype

Year of occupancy		1978
Geometry	Number of floors	10
	Height	30 m
	Surface area	295.9 m ²
	Number of apartments / floors	3 apartments / floor
	Volume	8877 m ³
Site altitude		11 m
Construction system		« PASCAL » heavy prefabrication process
Exterior wall	Components	Prefabricated sandwich panels of reinforced concrete + plastering + painting
	U-value [W·m ⁻² K ⁻¹]	2.30
Glazing	Components	Single glazing
	U-value [W·m ⁻² K ⁻¹]	6.00
Roofing	Components	Prefabricated sandwich panels of reinforced concrete + multi-layer bituminous waterproofing
	U-value [W·m ⁻² K ⁻¹]	1.20

Overview on the building energy data and system

The collection of real energy consumption data is required for the calibration and validation of the building's energy model. The energy consumption data were collected from the commercial agency located in Benhouria street affiliated to SONELGAS (National Electricity and Gas Distribution Company), in the form of

quarterly energy bills for all quarters of the year 2021, for 10 apartments, through sampling, choosing apartments of different floors heights, and different orientations, and then deducting the amount of energy consumed by the building as a whole. It's important to take into account that in Skikda, similarly to the rest of the Algerian country, natural gas is primarily used for heating systems, domestic hot water,

as well as cooking, whereas electricity is primarily used for lighting, cooling, and domestic devices (TV, dishwasher, computer, cooling systems, fridges and freezers and so on).

After the launch of renovation scenarios, once the PVT hybrid system has been integrated, its thermal production will be used to cover the natural gas demand (heating and hot water needs, through boilers and radiators supplied with energy from a PVT solar hybrid system), while its electrical production will replace electricity from the city's central grid. With the exception of the (P.R.6), where the hot water needs were provided par hybrid PVT electricity supply, for the use of the electrical boilers. The renewable energy produced in each of the six renovation scenarios is combined with the previous supply systems relying on fossil fuels to fulfill the buildings' requirements.

Materials

Several tools were used in this research, including QGIS tool, Meteonorm, city energy analyst. Where QGIS was used in this present research for the modeling of the case study and

its surrounding urban context, by creating a file (Shapefile) which contains all the necessary spatial and geographical data.

While Meteonorm was used to bring out the climatic data for the city of Skikda, in file format (EPW), which was imported into the CEA software, this last one uses this climatic data alongside the geospatial data provided by QGIS. Also, CEA adheres to the reduced-order UBEM approaches, which refers to Urban building energy modeling tools, it's commonly used for performing fast studies of building efficiency. This method requires a smaller amount of input in comparison to the physical as well as data-based methods (Hong et al., 2020). Only general information on the construction's geometry (including the number of floors as well as height), construction year along with renovation year(s), building system typology, and the principal building function are necessary to match the archetype data to the actual building (Willmann et al., 2019). It was used in this research to run monthly solar analysis, and to assess the initial state of the building, and then to highlight the effects of the interventions made in each of the scenarios launched.

Results and Discussion

Solar analysis

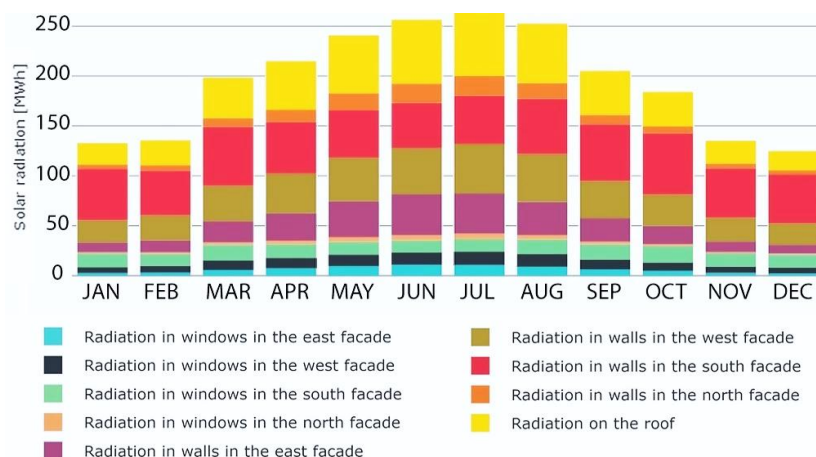


Fig. 2. Monthly solar energy irradiation on outdoor surfaces in 2021, simulated by CEA (By the authors).

After entering the building, climate and geospatial data and running the solar simulation, the monthly incident solar energy quantities on the roof and on the various vertical surfaces are shown in Figure 2.

The bar chart shows that radiation to the building envelope was highest in July, at around 270 MWh, and lowest in December, at 130 MWh. This shows the big difference between these two extreme periods in terms of

radiation and solar energy reception, since the amount received in December was less than half that received in July.

During the months with the highest temperatures (May, June, July, August), the greatest percentage of radiation is on the roof, whereas during the rest of the year, the greatest percentage of radiation is on the south façade, given that the angle of the sun's height is reduced during the winter period (Alberto, 2021), and that the surface of the vertical façades of the buildings is greater than the surface of the roof, due to the height of the building, which reaches almost 30 meters, in the absence of urban or natural obstacles, particularly on the south side ; which justifies the choice of scenarios involving the integration of photovoltaic-thermal panels on the roof and on the south facade.

Prior state energy and ecological analysis

The results of the energy model calibration were validated on an annual and monthly basis by comparing simulated and actual total energy consumption. For the archetype studied, the value of the CVRMSE coefficient (Coefficient of Variation of the Mean Square Error) is between 0 and 15, while the NMBE value (Normalized Mean Bias Error) is limited to between -5 and +5, as is the percentage error in the annual validation, which is close to 0. This means that the error percentages are acceptable and that

the energy models have been validated in accordance with the internationally accredited ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers) 14-2002 guidelines to meet the requirement for a standard set of methods for calculating energy savings (as well as demand). The objective of Directive 14-2002 is to provide a minimum acceptable level of performance in the measurement of energy and demand savings achieved through energy efficiency projects applied to residential, commercial or industrial buildings. (Haberl et al., 2005). After calibration and validation, bar chart of simulated monthly energy consumption was presented and plotted in (Figure 3).

The bar chart confirms that energy requirements during the cold period increase sharply, reaching almost 70 MWh in January, in contrast to the summer months, when consumption is only 10 to 15 MWh. As a result, the highest monthly energy consumption in summer is less than 5 to 7 times higher than in winter. Almost 70% of annual energy consumption was devoted to heating and 17% to hot water production (for cooking, washing and bathing); these two needs, which represent a very high percentage of annual energy consumption, were met by natural gas. The remaining needs are covered by the electricity network and include cooling, artificial lighting, electrical appliances, etc.

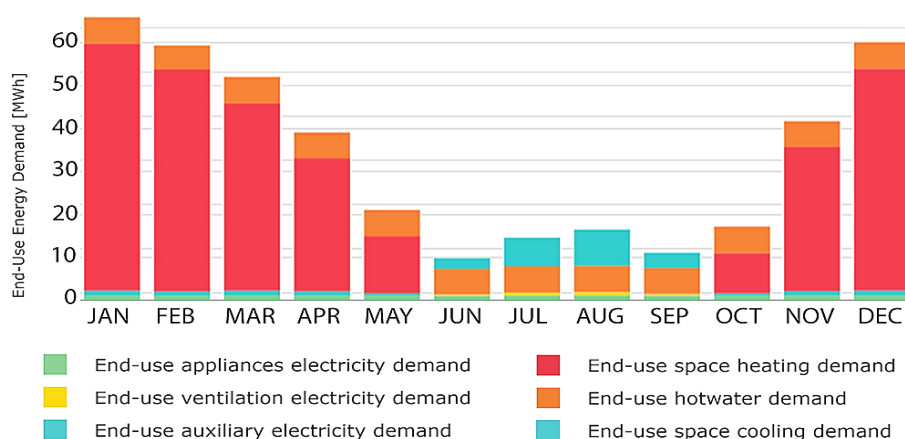


Fig. 3. Monthly energy consumption over the year 2021, before renovation, according to CEA tool simulations (By the authors).

For the energy and environmental assessment of the initial state (before renovation),

and due to the absence of an Algerian benchmark, this study used the new French

benchmark for the energy performance diagnosis (DPE) of housing in 2021 (Batiactu, 2022; Vlad, 2022), due to climatic convergence, since the context of the city of Skikda and the French state both belong to a bit similar Mediterranean climatic zone, according to the Köppen-Geiger climate classification (Chen, 2023).

Figure (4) shows the different energy and environmental classes, as well as the ranges of energy consumption (KWH/m²/year) and greenhouse gas emissions (kg Co₂/m²/year) for each class.

By comparing the building's annual energy consumption and the amount of carbon

emissions simulated by the CEA tool, for the year 2021, on the basis of the data and energy systems used. The annual consumption per unit area, which reached 138.37 KWH/m²/year, and the annual operational GHG emissions per unit area, which were 28.09 kg of CO₂/m², indicate that the energy consumption and carbon emissions of the residential building studied (baseline scenario) are in class C, which means that this building is an average energy consumer and carbon emitter (neither efficient, nor over-energetic, nor a high carbon emitter).

	Energy consumption	CO2 Emissions
A	≤70 Kwh/m ² /year	≤6 Kg. Co ² /m ² /year
B	71 to 110 Kwh/m ² /year	7 to 11 Kg. Co ² /m ² /year
C	110 to 180 Kwh/m ² /year	12 to 30 Kg. Co ² /m ² /year
D	181 to 250 Kwh/m ² /year	31 to 50 Kg. Co ² /m ² /year
E	251 to 330 Kwh/m ² /year	51 to 70 Kg. Co ² /m ² /year
F	331 to 420 Kwh/m ² /year	71 to 100 Kg. Co ² /m ² /year
G	≥ 420 Kwh/m ² /year	≥101 Kg. Co ² /m ² /year

French referentiel for
energy and environnemental
classification of buildings

Fig. 4. The French benchmark used in the energy and environmental assessment of the archetype studied (Source: by the authors, on the basis of (Batiactu, 2022; Vlad, 2022) data).

On the basis of considerations that 87% of the building's energy use is based on natural gas, and that also the building emitted 83.14 ton of operational CO₂ in the year 2021, and in accordance to the American Lung Association, that have declared that the combustion of natural gas generates also dioxide (NO₂), an atmospheric pollutant that can diminish lung function, promote airway inflammation, aggravate asthma episodes, and worsen other respiratory ailments.

Beside an increasing amount of research in the medical field has established a correlation between gas cooking together with childhood asthma, involving a 2013 meta-analysis which found that children who reside in dwellings

with gas stoves have a 42% higher probability of symptoms of asthma as well as a 24% grew risk of getting diagnosed with asthma over their lifetime. Another 2008 Johns Hopkins research discovered similar connections (Pearce, 2022). Energy renovation scenarios are suggested, to assess the influence of the interventions linked to each scenario on the reduction of energy consumption of gas and electricity in the building studied, in order to improve the energy as well as ecology state.

Energy and ecological analysis of post-renovation scenarios

Table (2) shows the annual energy requirements in gas and electricity, before and after

the different intervention scenarios, as well as the annual production of renewable energy after each scenario related to the integration of hybrid PVT system.

Table 2. Annual energy consumption and production for each scenario launched

Scenarios	Annual energy demand (Year 2021) (MWH/year)		Annual renewable energy production (Year 2021) (MWH/year)	
	Gaz	Electricity	Thermal production	Electricity
Baseline scenario	364.55	44.91	-	-
(P.R.1)	267.66	46.25	-	-
(P.R.2)	260.37	46.41	-	-
(P.R.3)	260.37	46.41	149.75	33.30
(P.R.4)	260.37	46.41	8.50	27.82
(P.R.5)	260.37	46.41	19.91	71.49
(P.R.6)	187.48	119.3	19.91	71.49

The scenario (P.R.1) reveals a 26.57% reduction in annual gas consumption, reflecting the ability of triple-glazed windows to enhance thermal insulation and reduce heat loss; on the other hand, there is an increasing of 1.34 MWH in energy consumption compared with the initial state; this can be justified by the requirement for more artificial lighting, due to the low light transmission of triple-glazed windows compared with single-glazed windows, according to the study (Ha-Nhi, 2023), where it is cited that the extra layers of glass and air between the glasses restrict the amount of light reaching the interior of the building. This type of glazing affects the quality of sunlight as well as reducing its quantity. The building immediately appears darker and gloomier.

Replacing (triple-glazed windows) with (triple-glazed windows combined with two selective low-emissivity coatings) in the (P.R.2) has shown that the latter have a slightly higher thermal insulation capacity than the former, given that the reduction in gas consumption for heating needs is reduced by up to 28.58% compared to the initial scenario (Baseline). On the other hand, there is a slight increase in electricity consumption, given that the latter type of glazing is darker, and requires much more artificial lighting.

In scenarios (PR3) and (PR4), the integration of the same surface area (192 m²) of photovoltaic-thermal panels, respectively on the roof and on the south-facing façade (on surfaces receiving an annual cumulative energy quantity greater than 1620 and 1190 [kWh/

m²/year], respectively, as maximum values of annual cumulative solar radiation on the roof and on the south-facing facade), showed that the panels in these two scenarios produced converging quantities of electrical energy, with a slight increase for the panels installed on the roof. In terms of thermal production, the roof-mounted panels produced up to 18 times more than the amount produced by the south-facing panels.

The integration of 344 m² of hybrid PVT panels on the south facade, in areas receiving an annual cumulative energy quantity greater than 1150 [kWh/ m²/yr.], in scenario (P.R.5) showed that despite the large surface area of this type of panel installed on the south facade, by adding 152 m² of panels compared with the two previous scenarios, thermal production remained well below that of scenario (P.R.3), and was not significantly increased compared with scenario (P.R.4).

On the other hand, electricity production increased sharply in scenario (P.R.5), more than twice that of scenario (P.R.3), and exceeded electricity requirements by almost 25 MWH, which necessitated a change in the supply system for certain types of use, and suggested the launch of scenario (P.R.6).

The comparative analysis between scenarios (P.R.3), (P.R.4) and (P.R.5) showed that the thermal production through the hybrid photovoltaic-thermal panels is linked to the annual cumulative solar radiation per unit area, over the year; whereas the electrical production is linked to the width of the panel surfaces.

Figure 5 shows the monthly production of renewable electrical and thermal energy for each of the scenarios (P.R.3), (P.R.4) and (P.R.5), using photovoltaic-thermal panels.

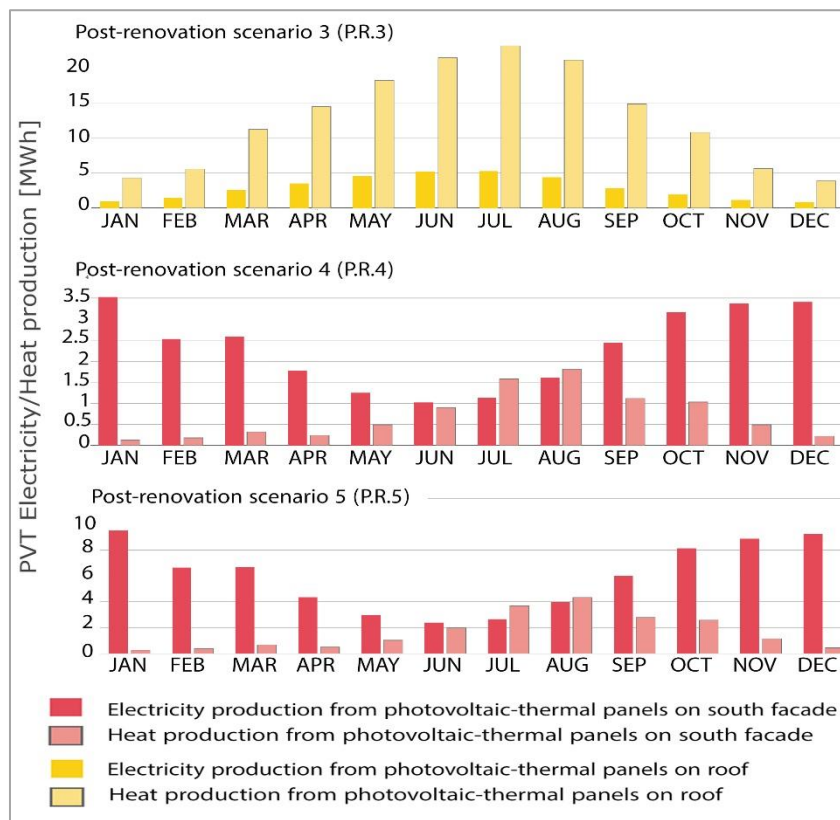


Fig. 5: The monthly electrical and thermal energy production for each of the scenarios (P.R.3), (P.R.4) and (P.R.5), simulated by CEA tool (By the authors).

For the scenario (P.R.3), it can be seen that the energy produced, whether thermal or electrical, increases during the hot summer period, with thermal production very high compared to electrical production, where thermal production goes up to 20 MWh during the months (June, July and August). On the other hand, both thermal and electrical production were reduced during the winter period, in particular thermal production, which did not even exceed 4.5 MWh during the coldest periods of the year (January and December).

But there was an inverse correlation between monthly thermal production and energy requirements for heating and hot water, since production was highest when thermal requirements were lowest, and vice versa; this calls for the use of seasonal thermal energy storage systems (STES), a method that could be, for example, a borehole thermal energy storage system (BTES), which is an underground structure

that stores huge quantities of solar heat captured in summer for use later in winter. Essentially, it's a huge underground heat exchanger. A BTES is made up of a series of boreholes that look like ordinary drilled wells. A polyethylene pipe with a "U" bend at the bottom is placed down the borehole after drilling. The borehole is subsequently plugged with an extremely heat-conductive grouting material to ensure good heat exchange with the surrounding earth (Gehlin, 2016).

There are many other storage systems, such as water-based storage, rock beds, while the cheapest in terms of economics among these three systems is borehole thermal energy storage (BTES), according to (Xu et al., 2014).

On the other hand, in both scenarios (P.R.4) and (P.R.5), where the panels are installed on the south facade, electricity production was significantly higher than thermal production, up to 3.5 MWh in scenario (P.R.4) and more

than 9 MWH in scenario (P.R.5). Electricity generation increases in winter and decreases in summer, while thermal generation increases in summer and decreases in winter, but to a very small extent compared to scenario (P.R.3).

Electricity production, which is higher in these two scenarios than thermal production, is therefore in direct correlation with the amount of monthly energy requirements, but as was

noted in scenario (P.R.5), production was higher than consumption, which suggests the launch of a sixth scenario to analyze the benefit of all this electrical energy, changing the natural gas-based hot water system for an electric one.

Figure (6) visualizes the graphical presentation of final fossil energy consumption, for each scenario.

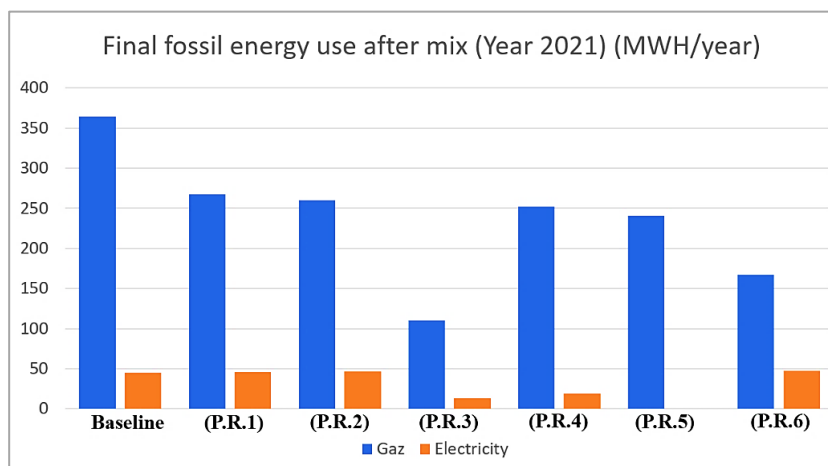


Fig. 6: The graphical presentation of the final energy use of energy from fossil sources for each scenario (By the authors).

According to this graphical presentation, the energy savings are between 26.57% and 28.58% in the cases (P.R.1) and (P.R.2), respectively, as previously indicated, so that there is only a very slight difference between the use of triple-glazed windows and the use of triple-glazed windows combined with two layers of selective low-emissivity coatings.

In scenario (P.R.3), there was a significant reduction in the need of gas and electricity from the fossil fuel supply network, resulting in total energy savings of 69.78% compared to the baseline and 41.2% compared to scenario (P.R.2), and is therefore the most efficient of the 6 scenarios. However, the installation of hybrid panels on the south facade did not result in significant energy savings, where total energy savings only increased by 5.37% and 12.69% in scenarios (P.R.4) and (P.R.5) respectively compared to scenario (P.R.2), despite the huge surface of solar panels in the (P.R.5).

The change to the hot water supply system, alongside the integration of 3448 m² of photovoltaic-thermal panels on the south-facing vertical surfaces and the replacement of single glazing with triple glazing combined with two selective low-emissivity coatings, in scenario (P.R.6), make this the second-best scenario in terms of energy savings, with an energy saving of 47.40% compared with the initial state.

Table (3) represents the building's energy and environmental rating for each scenario, compared with the new French DPE standard (Figure 5). The results disclose that the building has become energy class (B), and remains ecologically class (C), in all post-renovation scenarios (they have become energetically good, but ecologically stayed medium), with the exception of scenario (P.R.3) where the energy and environmental class has become (A), and has therefore become very high performer in terms of energy and ecology.

Table 3. Energy and environmental classification of the archetype according to each renovation scenario by comparison with the new French benchmark.

Scenarios	Total fossil energy use after mix (Year 2021) (MWH/year)	Total fossil energy use after mix (Year 2021) (KWH/m ² /year)	Energetic class	Total annual operational GHG emissions (ton of CO2)	Total annual operational GHG emissions (Kg of CO2/m2)	Environmental class
(P.R.1)	313.91	106.08	B	76.28	25.77	C
(P.R.2)	306.78	103.67	B	74.20	25.06	C
(P.R.3)	123.73	41.81	A	18.11	6.12	A
(P.R.4)	270.46	91.40	B	72.86	24.61	C
(P.R.5)	240.46	81.20	B	70.11	23.69	C
(P.R.6)	215.38	72.76	B	43.79	14.79	C

Scenario (P.R.3) was the most appropriate, due to the significant annual thermal productivity of the roof panels, in time where 87% of the building's annual needs are thermal energy-related, in addition to the enhanced insulation of the envelope through the use of highly efficient types of glazing in terms of thermal capacity, in order to reduce heating requirements; consequently, this combination resulted in a considerable reduction in both energy consumption and ecological footprint.

Conclusion

This study was carried out using a numerical method, based on real data measured throughout 2021, and the UBEM approach, in the context of the energy transition. The analysis of the initial state was maintained with a hybrid renovation through 6 scenarios, analyzing the energy and environmental aspects for each scenario, where the integration of hybrid PVT system was based on the solar analysis of the building which showed that the greatest amount of radiation received monthly was on the roof and on the south façade, and that the amount of annual solar radiation accumulated per unit area was much greater on the roof than on the south façade. Furthermore, the energy analysis of the initial state of the calibrated model of the building, which included 30 houses, can give an insight into the energy lifestyle of the inhabitants of Skikda, showing that energy requirements were very high in winter compared with other seasons, and that annual

heating and hot water requirements reached almost 87%, given that these two types of requirements are supplied by a system based on natural gas, which is very harmful to health and environment. Improving the thermal insulation of the building by replacing single glazing with triple glazing or triple glazing combined with two selective low-emissivity coatings can achieve significant annual energy savings of up to 26.57% and 28.58% respectively. While, the integration of hybrid PVT panels on the roof resulted in very high thermal production, but low electrical production, compared with integrating the same surface area of these panels on the south façade, or even doubling the surface area of these panels; this means that thermal production is mainly linked to the amount of solar radiation accumulated per unit area over the year; also noting that thermal production by roof-integrated panels was highest in summer and lowest in winter, which puts it in negative correlation with changes in energy requirements over the months of the year, and requires a seasonal thermal energy storage system, which must be associated with this scenario to take full advantage of thermal production. Thanks to the significant energy production of a 344 m² integration of photovoltaic-thermal panels, the switch from a natural gas-based energy supply system to an electric system contributed to energy savings and made the scenario the second-best, meaning that the right combination of improvement measures needs to be adopted.

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